

There are two MS/MS data files (unknownMSMS1 and unknownMSMS2, both in txt format) acquired from trypsin digested protein gel bands. The proteins were originally from *Arabidopsis thaliana*.

- 1) Please identify and report the two proteins from the attached data files using MS/MS ion search software from the links below.

http://www.matrixscience.com/cgi/search_form.pl?FORMVER=2&SEARCH=MIS

- 2) One of the proteins is phosphorylated, please determine which one and show evidence supporting the phosphorylation.
- 3) One of the samples is contaminated by human keratin. Which one is it, and how do you know?